

Nishant Sharma

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Full-stack Applied AI engineer with 3 years of experience building production web applications, backend APIs, and AI-powered workflows. Proficient in Python, TypeScript, Next.js, FastAPI, Node.js, PostgreSQL, Redis, Docker, and AWS. Built real-time LLM systems, customer-facing products, and reliability-focused cloud deployments.

Education

New York University, Tandon School of Engineering · New York, NY Master of Science in Computer Engineering · GPA: 3.7	May 2026
Dr. A.P.J. Abdul Kalam Technical University · Delhi, India Bachelor of Technology in Computer Science and Engineering · First Division with Distinction	Jun 2024

Technical Skills

Languages	Python · TypeScript · JavaScript · SQL · Java · C++
Frontend	React · Next.js · Tailwind · real-time UI
Backend & APIs	FastAPI · Node.js · Express · REST APIs · WebSockets · SSE · ARQ · Celery · Redis
AI Systems	LangGraph · RAG pipelines · LLM APIs · MCP servers · vector search · PyTorch · prompt engineering
Data & Storage	PostgreSQL · MySQL · MongoDB · Redis · Qdrant · AWS RDS
Cloud/Infra	AWS EC2 · S3 · RDS · SQS · Docker · Kubernetes · Terraform · GitHub Actions
Reliability	CI/CD · structured logging · rate limiting · async workers · integration testing · deployment automation

Projects

Singularity - AI Deep Research Platform · singularity.hellonish.dev	Mar 2026 – Apr 2026
- Built and deployed a full-stack AI research platform using Next.js, FastAPI, PostgreSQL, Redis, ARQ workers, and real-time SSE streaming. - Designed a DAG-based orchestration system coordinating 44 modular skills across 11 research domains, enabling long-running AI workflows from planning through synthesis. - Implemented production reliability features including Redis sorted-set rate limiting, orphan job recovery, fail-open behavior, and structured per-step execution logs. - Reduced embedding worker memory footprint by 94% by replacing PyTorch sentence-transformers with ONNX Runtime fastembed. - Added integration tests against real PostgreSQL and Redis containers in GitHub Actions, validating async job execution, auth flows, and streaming behavior before deployment.	
Wand - AI Career Intelligence Platform · github	Jan 2026 – Apr 2026
- Built an AI workflow platform that reduced manual company, JD, and contact research from 30–60 minutes to a single automated run. - Designed an asyncio DAG scheduler to execute independent pipeline steps in parallel, reducing wall-clock latency by 50%. - Streamed real-time pipeline progress from Celery workers to the browser using Redis pub/sub and WebSockets, eliminating client polling. - Routed tasks by model complexity and added profile caching, reducing per-analysis inference cost from \$0.10 to \$0.01 while saving 3,000 tokens per repeat run. - Added agent-level tracing for latency, token usage, request cost, structured output validation, and entity deduplication, making AI workflow behavior measurable across runs.	
PGDR - Persistent Gradient Disagreement Reweighting · paper	Jan 2026 – May 2026
- Built reproducible PyTorch evaluation pipelines across ERM, JTT, and GroupDRO baselines using identical seeds, training budgets, and model selection criteria. - Improved Waterbirds worst-group accuracy from 68.38% to 81.52% while preserving 92.56% average accuracy across 3 seeds.	

Work Experience

Co-Founder and Chief Technology Officer · Macverin Technologies · macverin.com	Jul 2022 – Aug 2023
- Delivered 12 client platforms in 14 months across ride-hailing, booking, visa consultancy, and internal operations domains using Node.js, FastAPI, React, Next.js, PostgreSQL, MySQL, MongoDB, and AWS. - Worked directly with clients to translate ambiguous workflows into shipped software, including intake, case tracking, scheduling, payments, and internal operations tools. - Built role-based application processing software for an 8-person visa operations team using Node.js, FastAPI, PostgreSQL, and Firebase Auth. - Integrated Stripe payments, Google Maps APIs, Firebase Auth, and AWS deployments across client platforms with isolated business logic per customer.	
Lead Software Engineer · Ingelt Study Abroad	Sep 2023 – Aug 2024
- Led a 3-engineer backend team, owning task delegation, code reviews, PR approvals, standups, and technical communication with company leadership. - Architected load-balanced AWS infrastructure with ALB, RDS, SQS, S3, and CloudFront, including async SQS pipelines and dead-letter handling for background jobs. - Built Jenkins CI/CD pipelines for automated test, build, and EC2 deployment, replacing manual releases with repeatable deployment workflows.	

Additional Experience

Machine Learning Teaching Assistant · NYU Tandon School of Engineering, New York, NY	Sep 2025 – May 2026
- Led debugging sessions for graduate students on PyTorch training pipelines, covering optimization, autograd, data loading, and reproducibility issues. - Standardized PyTorch assignment templates and troubleshooting guides, reducing environment-related debugging overhead during office hours.	